Kamyab Safaei Baghbaderani

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RESEARCH INTERESTS

Molecular Dynamics Simulations; Bioinformatics and computational biology; machine learning for biology and medicine; protein language models; kinase signaling; sequence–function prediction; quantitative cell biology; structural and computational biology; protein structure and dynamics; AI-assisted biomedical discovery.

EDUCATION

Sharif University of Technology, Iran Graduated: Summer 2024
Bachelor of Science, Chemical Engineering GPA: 16.20/20
Bachelor Thesis: Mechanics of Neutrophil Phagocytosis Grade: 19.7/20

Computational modeling of neutrophil phagocytosis for spherical and non-spherical targets under the supervision of Prof. Hanieh Safari.

STANDARDIZED TESTS

GRE General Test – Quantitative Reasoning **167/170**
TOEFL iBT 89

PROFESSION & EXPERIENCE

Raha Pharmaceutical Company Summer of 2023 and 2026
Laboratory Intern 480 hours

- Rotated through raw-material, finished-product, quality-control, microbiology, and water-quality laboratories, gaining practical exposure to pharmaceutical laboratory procedures, microbiological workflows, analytical quality control, and regulated production environments.

Programming and Scientific Software Development Freelance / Independent

- Developed Python/Django applications for microbiology, data analysis, scientific computation, and research visualization, including bacterial colony analysis and research dashboards.

TECHNICAL SKILLS

- **Computational Biology Machine Learning:** Bioinformatics, protein language models (pLMs), ESM2 protein embeddings, biological sequence representation, kinase–phosphosite prediction, zero-shot/generalized zero-shot learning (ZSL/GZSL), PyTorch, scikit-learn, GPU/CUDA computing, model evaluation, and multi-seed experimentation.
- **Scientific Computing Data Analysis:** Python, NumPy, SciPy, Pandas, MATLAB, COMSOL, numerical modeling, statistical analysis, data preprocessing, scientific visualization, Matplotlib, Plotly, and computational simulation.

- **AI, RAG Research Software:** Retrieval-Augmented Generation (RAG), vector databases (ChromaDB), embeddings, LLM APIs, agentic AI and tool-using systems, Django, REST APIs, SQLite, Git/GitHub, pytest, experiment tracking, reproducibility, validation pipelines, and research-oriented software development.
- **Optimization Engineering Modeling:** GAMS, mathematical optimization, objective-function formulation, Aspen HYSYS, Simulink, process simulation, parameter analysis, and engineering-system modeling.

PROJECTS

DARKIN LabOS – pLMs for Kinase–Phosphosite Prediction

Repository

Built DARKIN LabOS, a reproducible computational-biology and machine-learning research workspace extending the DARKIN kinase–phosphosite benchmark, inspired by Prof. Öznur Taştan and colleagues' work on protein language models for kinase prediction. Processed and validated 12,093 ESM2-650M protein embeddings and implemented multi-seed zero-shot and generalized zero-shot learning (ZSL/GZSL) experiments on CUDA, including deterministic training, model evaluation, and reproducibility checks. A semantic extension improved five-seed ZSL macro AP from 0.0926 to 0.1198, strengthening my experience in bioinformatics, protein representation learning, and machine learning for understudied kinase classes.

Mechanics of Neutrophil Phagocytosis – Bachelor Thesis

Repository

Developed a computational model to investigate neutrophil phagocytosis and the interaction of cells with spherical and non-spherical targets. Applied numerical modeling, differential-equation solving, chemotaxis and adhesion modeling, and scientific visualization using Python, NumPy, SciPy, and Matplotlib. The project combined engineering mechanics with quantitative cell biology and strengthened my interest in computational approaches to biological systems; thesis grade: 19.7/20.

Colony Counter DNA Primer Suggestion – Microbiology Software

Repository

Developed a Django-based microbiology application for automated bacterial colony counting and DNA-primer suggestion. Implemented image-processing workflows using Python, NumPy, scikit-image, and Matplotlib, together with DNA-sequence input, forward/reverse primer suggestions, melting-temperature and GC-content calculations, and interactive result visualization. The project combined microbiology, biological data processing, image analysis, and scientific software development in a practical research-oriented tool.

LANGUAGES

Persian	Native
English	TOEFL iBT: 89
Italian	A2, currently learning
French	A2

HOBBIES

Basketball	
Painting	Digital and Acrylic
Violin	Since 2016